KENYA MEDICAL TRAINING COLLEGE
FACULTY OF REHABILITATIVE HEALTH SCIENCES
DEPARTMENT OF MEDICAL ENGINEERING

FINAL QUALIFYING EXAMINATION
FOR
CERTIFICATE IN MEDICAL ENGINEERING TECHNOLOGY

PAPER: ELECTRICAL ENGINEERING TECHNOLOGY

DATE: 3rd February, 2023 **TIME: 3 HOURS (9:00 A.M – 12 NOON)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. All multiple choice questions **MUST** be answered in **CAPITAL LETTERS**
4. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.
5. Ensure that all examination answer scripts are handed in at the end of the examination
6. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

ELECTRICAL ENGINEERING TECHNOLOGY

SECTION 1: MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. Electromagnetic energy conversion devices are built with air gap in the magnetic circuit to –
   A. Increase reluctance of magnetic path
   B. Decrease reluctance of magnetic path
   C. Separate the fixed and moving parts
   D. Strengthen cross-magnetization
2. Which of the following contributes the greatest heat loss in a d.c machine?
   A. Copper losses
   B. Eddy current losses
   C. Frictional losses
   D. Hysteresis losses
3. What is efficiency of a DC generator?
   A. Electrical efficiency
   B. Overall efficiency
   C. Mechanical efficiency
   D. All of the above
4. Inter- pole windings are connected_____
   A. In parallel with armature winding
   B. Partly in series and partly in parallel with armature winding
   C. In series with armature winding
   D. In series with the field winding
5. What is the effect of armature reaction to the flux under a leading pole tip?
   A. Is decreased
   B. Is increased
   C. May decrease or increase
   D. Remains unchanged
6. The coil undergoing commutation _____
   A. Remains open circuited by the brush
   B. Remains short circuited by the brush
   C. Remains neutral
   D. Remains stack
7. Which type of loss in a transformer is equal to the no load input power?
   A. Eddy current
   B. Copper
   C. Iron
   D. Frictional

8. The voltage per turn of the primary of a transformer is
   A. Twice the voltage per turn of the secondary
   B. More than the voltage per turn of the secondary
   C. Less than the voltage per turn of the secondary
   D. Same as the voltage per turn of the secondary
9. The efficiency of a transformer is zero at _____
   A. Full – load
   B. No – load
   C. Half full –load
   D. Short circuit
10. Rotor windings of a 3-phase wound rotor is connected in _____
   A. Star
   B. Delta
   C. Partly star and partly delta
   D. Zigzag
11. An induction motor is preferred to a.D.C motor because it_____
   A. Provides high starting torque
   B. Provides fine speed control
   C. Has a simple and rugged construction
   D. All of the above
12. If the slip of a 3 phase induction motor increases, power factor of the rotor _____
   A. Is increased
   B. Is decreased
   C. Remains unchanged
   D. Is zero
13. An induction motor on load resembles a transformer whose secondary is_____
   A. Open circuited
   B. Supplying available resistive load
   C. Short –circuited
   D. Excited
14. Which type of motor is commonly used in laundry machines?
   A. Single phase series
   B. Split - phase
   C. Shaded pole
   D. Hysteresis
15. The number of poles in the start winding of an induction motor are _____
   A. Same as the number of poles in the main winding.
   B. Less than the number of poles in the main winding
   C. More than the number of poles in the main winding
   D. More or less the number of poles in the main winding

16. A capacitor-start, capacitor - run motor has
    A. Low power factor
    B. High power factor
    C. High efficiency
    D. High starting torque
17. The frequency of E.M.F generated in an 8-pole alternator running at 900 RPM is _____
    A. 50 Hz
    B. 120 Hz
    C. 60 Hz
    D. 150 Hz
18. High speed alternators are driven by
    A. Diesel
    B. Hydraulic turbines
    C. Steam turbines
    D. Wind turbines
19. The examples of continuous energy conversion devices are:
    A. Contractors
    B. Relays
    C. Solenoids
    D. Motors and generators
20. Conductors used in compensated windings are housed_____
    A. In the slots in the pole faces
    B. In the slots in the armature
    C. In the slots in the stator
    D. All the above
21. Which of the following is not a method of controlling the speed of a. D.C motor?
    A. Varying the flux per pole
    B. Varying the number of windings in the armature
    C. Varying the resistance in the armature
    D. Varying the applied voltage
22. Why are motors starters important?
    A. Accelerating the motor
    B. Starting the motor
    C. Avoiding excessive starting current
    D. Preventing dust from blowing
23. Which of the following is not a method of generating E.M.F in a conductor?
    A. Self excitation
    B. Self induction
    C. Mutual induction
    D. Induction by motion

24. Which of the following is a method changing the emf induced is a coil?
    A. Varying the flux density
    B. Varying active length of coil
    C. Varying the speed at which the coils cuts the flux
    D. All of the above
25. Which of the following losses does not occur in a transformer?
    A. Iron losses
    B. Mechanical losses
    C. Copper losses
    D. Eddy current losses
26. What is the purpose of a load in an electrical circuit?
    A. To increase the circuit current
    B. To decrease the circuit current
    C. To utilize electrical energy
    D. To make the circuit complete
27. What is ratio between reactive power and apparent power of an ac load?
    A. Quality factor
    B. Reactive power
    C. Power ratio
    D. Power Factor
28. Why is a separately exited generator not used ?
    A. It is costly
    B. A separately d.c source is required for the field current
    C. Voltage drops considerably with load
    D. It is bulky
29. In a D.C machine, the number of commutator segments is equal to_____
    A. Number of conductors
    B. Number of coils
    C. Twice the number of poles
    D. Twice the number of coils
30. In an A.C circuit, power dissipated as heat depends on_____
    A. Impedance
    B. Capacitive reactance
    C. Resistance
    D. Inductive reactance
31. Which of the following does not generally affect the value of a capacitor?
    A. The dielectric material
    B. The surface area of the plates
    C. The thickness of the dielectric
    D. The voltage applied to the plate

32. In order to have a sparkles commutation, brushes should be placed _____
    A. A long the G.N.A
    B. Along the M.N.A
    C. Along G.N.A or M.N.A
    D. Parallel with the G.N.A
33. Flux distribution due to armature reaction causes the M.N.A in a generator to_____
    A. Move in the direction of rotation
    B. Remain stationery
    C. Move opposite to the direction of rotation
    D. Coincide with the G.N.A
34. The starting torque of a squirrel cage induction motor is_____
    A. Low
    B. Negligible
    C. Same as full-load torque
    D. Slightly more than full-load torque
35. What is the function of the iron core in a transformer?
    A. To provide support to the windings
    B. Reduce hysteresis losses
    C. Decrease the reluctance of the magnetic path
    D. Reduce Eddy current losses
36. What will happen to a 3-phase induction motor if there is a phase failure in one of the phases when the is motor running?
    A. It will slow down in speed
    B. It will stall
    C. It will increase the torque
    D. It will draw more current and may burn out
37. The strength of an electro-magnet can be increased _____
    A. By switch on and off the current
    B. By increasing the length of the conductor
    C. By increasing the current
    D. By minimizing magnetic flux leakage
38. Which among the following does not affect the resistance of a conductor?
    A. Temperature
    B. The length of the conductor
    C. The thickness of the material
    D. The shape of the conductor

39. According to Faraday's law of electro-magnetic induction, an E.M.F is induced in a conductor whenever it _____
    A. Lies in a magnetic field
    B. Cut the magnetic flux
    C. Moves parallel to the direction of the magnetic field
    D. Lies perpendicular to the magnetic field
40. The most efficient form of damping employed in electrical measuring instruments is
    A. Air friction
    B. Fluid friction
    C. Eddy current
    D. All of the above

SECTION 2: SHORT ANSWER QUESTIONS (40 MARKS)

41. List five (5) possible causes of failure in starting a D.C motor **(5 marks)**
42. State three (3) advantages and two disadvantages (2) of three phase transformers over singe phase transformers? **(5 marks)**
43. i) What is slip as used in a 3-phase induction motor? **(2 marks)**

    ii) A 6-pole, 3-phase induction motor is connected to a 50Hz supply. If it is running at 970 R.P.M, calculate the slip. **(3 marks)**
44. State three (3) advantages and two (2) disadvantages of separately excited generator. **(5 marks)**
45. i) Define a transformer? **(1 mark)**

    ii) State four (4) important applications of a transformer. **(4 marks)**
46. i) State three (3) main causes of over current in a three-phase induction motor. **(3 marks)**

    ii) What are the three (3) methods used to start a 3-phase induction motor. **(2 marks)**
47. i) State three (3) causes of poor commutation in D.C machines. **(3 marks)**

    ii) Name two (2) methods of improving commutation in D.C machines. **(2 marks)**
48. Outline five (5) cooling methods used in a transformer. **(5 marks)**

SECTION 3: LONG ANSWER QUESTIONS (20 MARKS)

49. a) A shunt generator supplies 96 A at a terminal voltage of 200 V. The armature and field shunt resistance is 1.0Ω and 50Ω respectively. The iron and frictional losses are 2500W. Find:
    (i)E.M.F generated **(2 marks)**
    (ii)copper losses **(4 marks)**
    (iii)commercial efficiency **(4 marks)**

b) i) State four (4) characteristics of transformers. **(4 marks)**

ii) An ideal 25KVA transformer has 500 turns on the primary winding and 40 turns on the secondary winding. The primary connected to 3000V, 50 Hz supply. Calculate:

a) Primary and secondary currents on full-load **(2 marks)**
b) Secondary E.M.F **(2 marks)**
c) The maximum core flux **(2 marks)**